Teleconnections

Basic Theory: Rossby Waves, Ray Tracing and Stationary Waves

Geoffrey K. Vallis

University of Exeter

Beijing, September 2019

http://tiny.cc/Vallis

Outline

Lecture 1

The Basics

- Teleconnections – what are they?
- Patterns of Variability
- Rossby Waves, Stationary Waves, Ray Tracing

Lecture 2

Some Current Research

- Numerical experiments.
- Teleconnections from the Tropics
- The North Atlantic Oscillation
  and Annular Modes

Teleconnections

What are they?

- Teleconnections are remote connections between different parts of the world that affect the climate and weather, typically on weekly to seasonal timescales.
- Wikipedia says:
  "Teleconnection in atmospheric science refers to climate anomalies being related to each other at large distances (typically thousands of kilometers). The most emblematic teleconnection is that linking sea-level pressure at Tahiti and Darwin, Australia, which defines the Southern Oscillation."

Two General Kinds

1. Large-Scale Oscillations and Patterns (e.g., 'see-saws')
   - El Niño and Southern Oscillation
   - Northern Annular Mode (NAM), SAM, NAO, Southern Oscillation, Pacific Decadal Oscillation, etc,
2. Remote Influences of Localized Events
   - Tropical SST Anomalies influencing North America and Europe
   - El Niño influencing monsoons.
   - etc

Applications and Examples

- Sea surface temperature anomalies and weather forecasting. Sea-surface temperature anomalies may last weeks to months. If they affect the weather remotely it gives us extra predictability.
- Atmospheric Oscillations: NAO, PDO, NAM, SAM are teleconnections. El Niño is a teleconnection (Tahiti to Darwin pressure anti-correlation).
- Do blocking events have teleconnections? Teleconnections *from* monsoons?
- Stratosphere tends to have long timescales. Tropospheric influence? A 'stratospheric bridge'?

A Few Early Papers up to 1980s

Gilbert Walker, 1923: Correlation in Seasonal Variations of Weather, VIII: A Preliminary Study of World Weather. *Memoirs of Indian Meteorological Dept.*

Jerome Namias, 1950s. Seasonal weather forecaster for NOAA. Published books of teleconnections for making long-term weather predictions.

Jacob Bjerknes, 1969: Atmospheric Teleconnections from the Equatorial Pacific. *Mon. Wea. Review.*

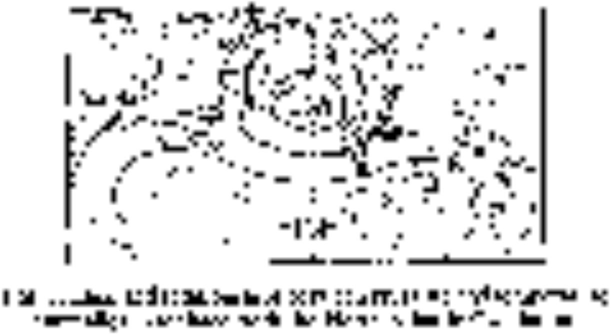
Wallace and Gutzler, 1981: Teleconnections in the geopotential height field during the Northern Hemisphere winter. *Mon. Wea. Review*

Hoskins and Karoly, 1981: The steady linear response of a spherical atmosphere to thermal and orographic forcing. *J. Atmos. Sci.*

Walker's Southern Oscillation

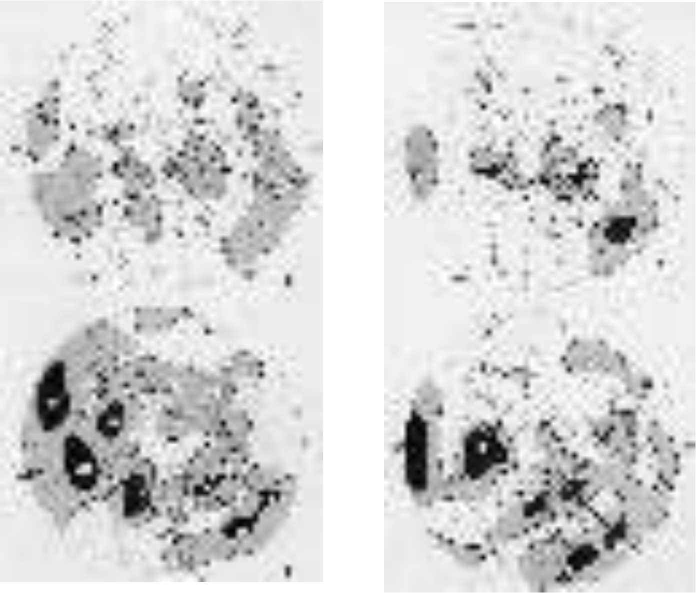
Pressure Correlations with Djarkata, Indonesia.
Berlage (1957) and Bjerkner (1969).

The NAO



Wallace and Gutzler, 1981

Northern Hemisphere



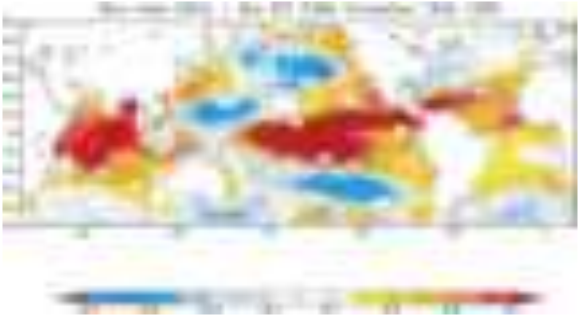
Strongest negative correlations
(in two datasets, 1962–1977 winter)

Sea-level pressure

500 hPa height.

Wallace & Gutzler, 1981.

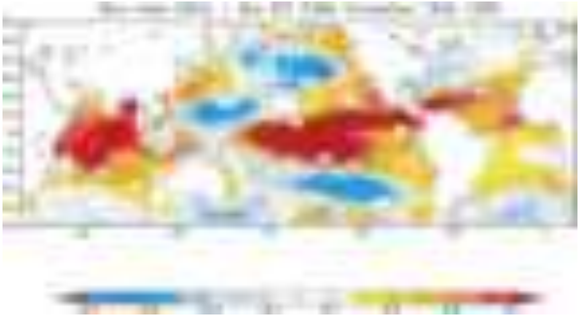
ENSO Influence



Correlation between El Niño SST anomalies and Global SST anomalies, 1950–1999.

Alexander et al, 2002.

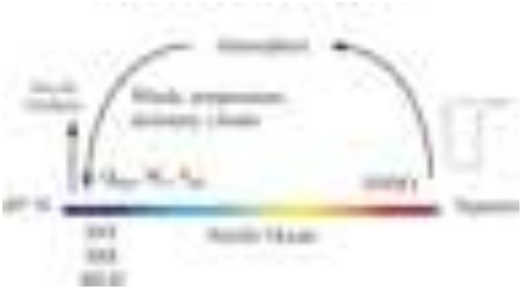
ENSO Influence



Correlation between El Niño SST anomalies and Global SST anomalies, 1950–1999.

Alexander et al, 2002.

The 'Stratospheric Bridge'



Sensitivity of N. Pacific Height Anomalies to SST

Reverse Engineer the SST Influence

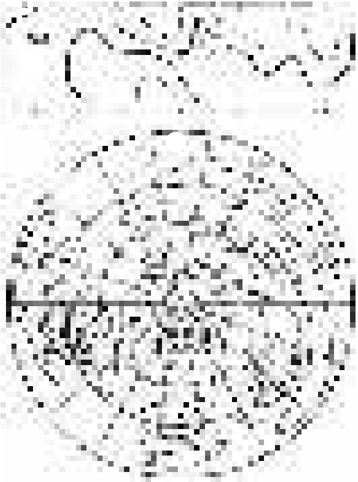
FIG. 15. Sensitivity map, or influence function, of a negative height anomaly over the North Pacific to anomalous SSTs 60 days earlier in the MLM. The influence function is contoured at an interval of 0.002 K m^{-1}; the zero contour is removed for clarity and positive (negative) values are red (blue). The target height response at 200 mb is indicated by gray shading with a 0.25-m interval.

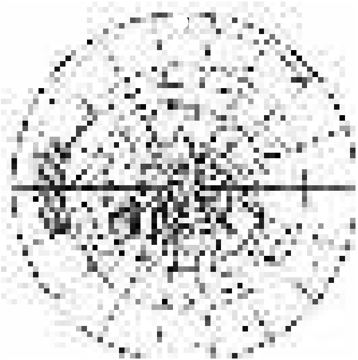
Influence function, or sensitivity map, of North Pacific height anomalies (grey shading) to SST anomalies (contours) 60 days earlier.

Alexander et al, 2002.

Teleconnection Examples

Link from the tropical sea-surface temperature anomalies to the mid-latitudes



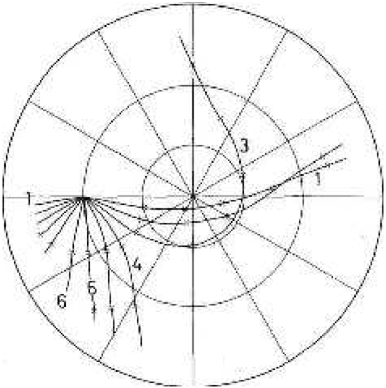


Hoskins and Karoly 1981.

Rays

A theoretical calculation of propagation

Source at 15°, differening background states.



A source in the tropics can have influence worldwide!

Hoskins and Karoly 1981.

Primitive Equations of Motion

Horizontal momentum:

$$\frac{\mathrm{D}u}{\mathrm{D}t} - fv = -\frac{1}{\rho}\frac{\partial p}{\partial x}, \qquad \frac{\mathrm{D}v}{\mathrm{D}t} + fu = -\frac{1}{\rho}\frac{\partial p}{\partial y}.$$

Hydrostatic:

$$\frac{\partial p}{\partial z} = -\rho g$$

Mass Conservation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \boldsymbol{v}) = 0$$

Thermodynamic:

$$\frac{\mathrm{D}\theta}{\mathrm{D}t} = Q$$

Equation of state

$$p = \rho R T.$$

On the beta-plane: $f = f_0 + \beta y$.

Vorticity and Potential Vorticity

Vorticity Equation:

$$\frac{\mathrm{D}\boldsymbol{\omega}}{\mathrm{D}t}_{a} = (\boldsymbol{\omega}_a \cdot \nabla)\boldsymbol{v} - \boldsymbol{\omega}_a \nabla \cdot \boldsymbol{v} + \frac{1}{\rho^2}(\nabla\rho \times \nabla p) \tag{1}$$

where $\boldsymbol{\omega}_a = \boldsymbol{\omega} + \boldsymbol{f}$.

Potential Vorticity:

$$\frac{\mathrm{D}Q}{\mathrm{D}t} \equiv \frac{\mathrm{D}}{\mathrm{D}t}\left(\frac{\boldsymbol{\omega}_a \cdot \nabla\theta}{\rho}\right) = 0. \tag{2}$$

Vertical component vorticity equation (approximate):

$$\frac{\mathrm{D}}{\mathrm{D}t}(\zeta + f) = -(\zeta + f)\left(\frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}\right) + \left(\frac{\partial u}{\partial z}\frac{\partial w}{\partial y} - \frac{\partial v}{\partial z}\frac{\partial w}{\partial x}\right) \tag{3}$$

Quasi-Geostrophic theory

Three main assumptions:

1. Small Rossby number: $\dfrac{U}{fL} \ll 1$.
2. Scale similar to Rossby deformation scale: $L \sim \dfrac{NH}{f}$.
3. Variations in Coriolis parameter are small (cf. beta plane).

Potential Vorticity Equation becomes:

$$\boxed{\frac{\mathrm{D}q}{\mathrm{D}t} = 0, \qquad q = \zeta + f + \frac{\partial}{\partial z}\left(\frac{f_0 b}{N^2}\right).}$$

$$u = -\frac{\partial \psi}{\partial x}, \quad v = \frac{\partial \psi}{\partial x}, \qquad \zeta = \nabla^2 \psi + \beta y \qquad b = f_0 \frac{\partial \psi}{\partial z}.$$

Perhaps the most important equation in dynamical meteorology!

Barotropic (One-layer) QG Equation

Much of Rossby-wave and teleconnection theory is based on the single-layer (no vertical variation) equations:

3D equation:

$$\frac{\mathrm{D}q}{\mathrm{D}t} = 0, \qquad q = \zeta + f + \frac{\partial}{\partial z}\left(\frac{f_0 b}{N^2}\right). \tag{4}$$

becomes

$$\boxed{\frac{\mathrm{D}q}{\mathrm{D}t} = 0, \qquad q = \zeta + f = \nabla^2\psi + \beta y.}$$

$$u = -\frac{\partial\psi}{\partial x}, \quad v = \frac{\partial\psi}{\partial x}, \qquad \zeta = \nabla^2\psi + \beta y \tag{5}$$

Rossby Waves

Begin with:

$$\frac{\partial \zeta}{\partial t} + \boldsymbol{u} \cdot \nabla \zeta + \beta v = 0.$$

Linearize about a mean flow.

$$\psi = \Psi(y) + \psi'(x, y, t),$$

where $\Psi = -Uy$. Gives:

$$\frac{\partial}{\partial t}\nabla^2\psi' + U\frac{\partial \nabla^2 \psi'}{\partial x} + \beta\frac{\partial \psi'}{\partial x} = 0.$$

Seek solutions

$$\psi' = \mathrm{Re}\,\widetilde{\psi}\,\mathrm{e}^{\mathrm{i}(kx+ly-\omega t)},$$

and obtain:

$$[(-\omega + Uk)(-(k^2 + l^2) + \beta k]\widetilde{\psi} = 0,$$

Gives *dispersion relation:*

$$\boxed{\omega = Uk - \frac{\beta k}{k^2 + l^2}\,,} \tag{6}$$

Group Velocity and Phase Speed

Phase speed:

$$c_p^x = \frac{\omega}{k} = U - \frac{\beta}{k^2 + l^2}.$$

Group velocity:

$$c_g^x = U + \frac{\beta(k^2 - l^2)}{\left(k^2 + l^2\right)^2}, \qquad c_g^y = \frac{2\beta k l}{\left(k^2 + l^2\right)^2}.$$

The wave activity (and sometimes the energy) in a Rossby wave propagates at the group speed:

$$\boxed{\frac{\partial \mathcal{A}}{\partial t} + \nabla \cdot \boldsymbol{c}_g \mathcal{A} = 0.} \tag{7}$$

This is one way teleconnections are made!

Rossby Waves in a Varying Medium

The Earth is not homogeneous! Even without topography.

Rossby wave equation:

$$\left(\frac{\partial}{\partial t} + U(y)\frac{\partial}{\partial x}\right)\zeta + v\beta^* = 0, \qquad (\beta^* = \beta - U_{yy}) \tag{8}$$

Dispersion relation:

$$\omega \equiv ck = U(y)k - \frac{\beta^* k}{k^2 + l^2}, \tag{9}$$

Seek solutions:

$$\frac{\partial^2 \widetilde{\psi}}{\partial y^2} + l^2(y)\widetilde{\psi} = 0, \quad \text{where} \quad l^2(y) = \frac{\beta^*}{U - c} - k^2. \tag{10a,b}$$

For the experts, the WKB solution is:

$$\widetilde{\psi}(y) = A_0 l^{-1/2} \exp\left(\mathrm{i}\int l \,\mathrm{d}y\right), \tag{11}$$

Group velocity:

$$c_g^x = U + \frac{(k^2 - l^2)\beta^*}{(k^2 + l^2)^2}, \qquad c_g^y = \frac{2kl\beta^*}{(k^2 + l^2)^2}, \tag{12a,b}$$

Turning Surfaces and Critical Surfaces

Meridional Wavenumber:

$$l^2(y) = \frac{\beta^*}{U - c} - k^2.$$

But suppose $U = c$? Or $l = 0$?

Two regions will stop propagation:

1. *A turning surface*, where $l^2 = 0$ and $c_g^y = 0$,
2. *A critical surface*, where $U = c$ and l^2 becomes infinite, and $c_g^y = 0$.

Condition for wave propagation is that:

$$0 < U - c < \frac{\beta^*}{k^2}.$$

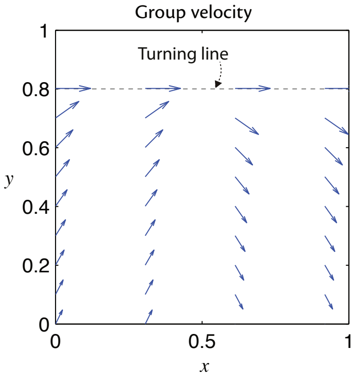
cf. Charney–Drazin condition.

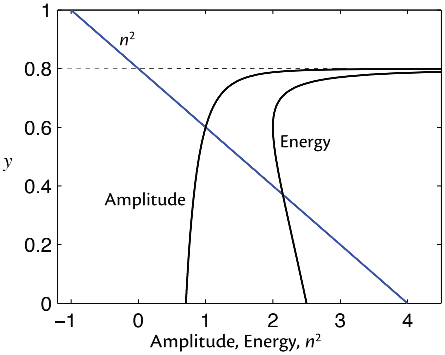
A Turning Line

A turning line, where $l^2 = 0$. Group velocity becomes zonal.

$$c_g^x - U = \beta^*/k^2 \neq 0, \qquad c_g^y \to 0.$$

Group velocity is purely zonal meridional wavenumber is large. The wave will *turn*.





$(n = l)$

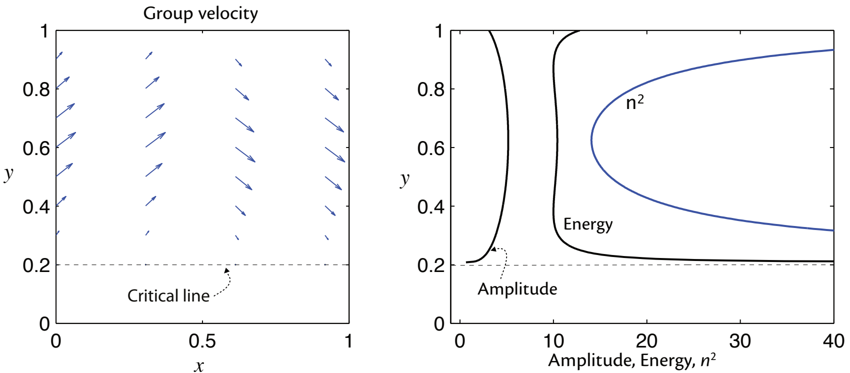
Amplitude ($\propto l^{-1/2}$) is infinite (in WKB calculation).

A Critical Line

A critical line, where $c = U$ and l is large. The wave stops!

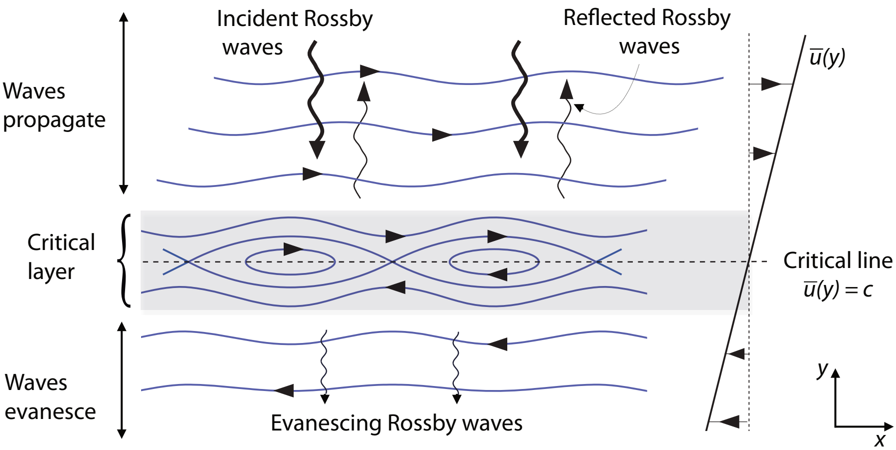
$$c_g^x - U \to 0, \qquad c_g^y \to 0, \qquad \frac{c_g^x - \overline{u}}{c_g^y} \to -\frac{l}{k} \to -\infty.$$

Group velocity is both directions is zero. The wave will break!



Critical Layers

In fact WKB theory breaks down and we get critical layers of finite thickness.



Ray Theory

- Ray theory tells us where waves go in inhomogeneous media, like the atmosphere.
- Similar approximations to those in WKB theory. Main one is:
  - Medium varies slowly compared to wavelength of wave.
  - Not well satisfied in atmosphere but proceed anyway!

Assume a solution of the form

$$\psi(\boldsymbol{x}, t) = a(\boldsymbol{x}, t)\mathrm{e}^{\mathrm{i}\theta(\boldsymbol{x},t)} \approx a(\boldsymbol{x}, t)\mathrm{e}^{\mathrm{i}(\boldsymbol{k}\cdot\boldsymbol{x}-\omega t)}.$$

with dispersion relation of the form

$$\omega = \Omega(\boldsymbol{k}; \boldsymbol{x}, t),$$

The wavenumber and frequency are functions of space and (in general) time.

After a bit of theory we find:

$$\frac{\partial \boldsymbol{k}}{\partial t} + \boldsymbol{c}_g \cdot \nabla \boldsymbol{k} = -\nabla\Omega,$$

and

$$\frac{\partial \omega}{\partial t} + \boldsymbol{c}_g \cdot \nabla\omega = \frac{\partial \Omega}{\partial t} = 0.$$

So frequency is constant along a ray.

Ray Theory in Practice

We follow a wave packet using the ray equations:

$$\frac{\mathrm{d}\boldsymbol{k}}{\mathrm{d}t} = -\nabla\Omega, \qquad \frac{\mathrm{d}\omega}{\mathrm{d}t} = 0, \qquad \frac{\mathrm{d}\boldsymbol{x}}{\mathrm{d}t} = \boldsymbol{c}_g\,. \tag{1}$$

Recipe

(1) Start with the location of the disturbance, and choose a wavenumber of interest (e.g. stationary wave, with $\omega = 0$).

(2) Calculate the group velocity.

(3) Move a short distance along in the group-velocity direction.

(4) Re-calculate the x and y wavenumbers using (1a). The frequency stays the same.

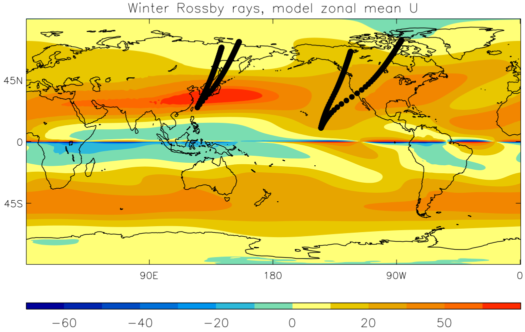
(5) Recalculate the group velocity.

(6) Go to item (3).

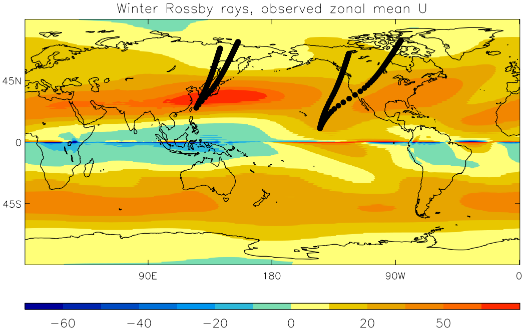
Examples

Use zonally-averaged zonal wind

Model



Observations

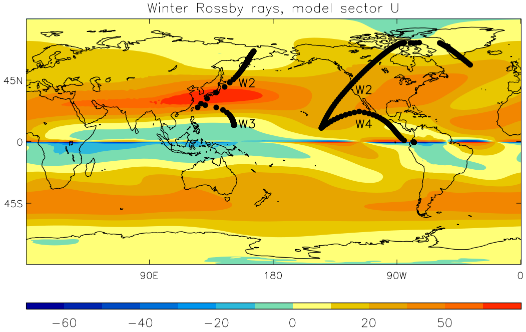


A. Scaife, UK Met Office

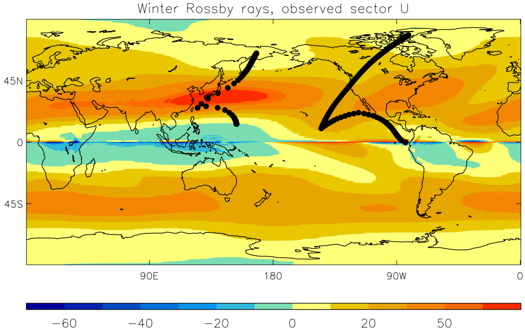
Examples

Use sector-averaged zonal wind

Model

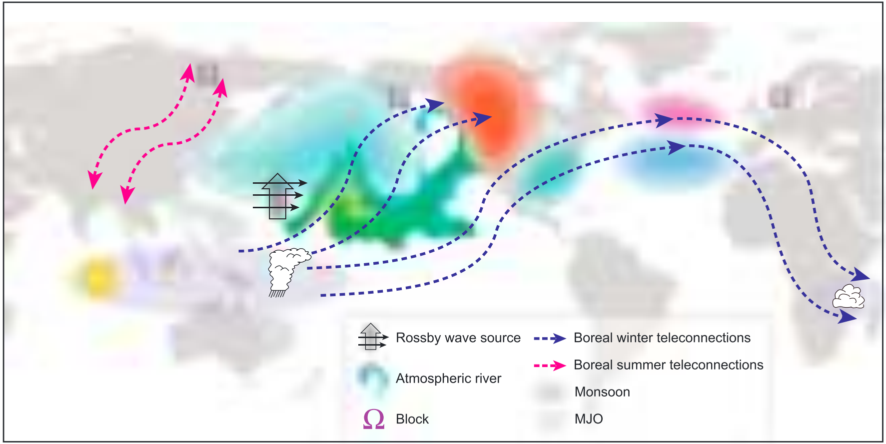


Observations



A. Scaife, UK Met Office

Teleconnection Schematic



Tropical–Northern Hemisphere interactions and teleconnections inferred from observational and numerical studies.

Stan et al 2017

Theory of Stationary Waves

- Stationary (Rossby) waves are a major factor in the large-scale zonally-asymmetric circulation.
- Due to orographic and thermal features at the surface of the planet.
- Understand using QG theory.

One-layer, orographic case

$$\frac{\mathrm{D}q}{\mathrm{D}t} = 0, \qquad q = \zeta + \beta y - \frac{f_0}{H}(\eta - h_b), \tag{13}$$

Linear version:

$$\frac{\partial q'}{\partial t} + U\frac{\partial q'}{\partial x} + v'\frac{\partial \overline{q}}{\partial y} = 0, \qquad \text{or} \qquad \frac{\partial}{\partial t}\left(\zeta' - \frac{\psi'}{L_d^2}\right) + U\frac{\partial \zeta'}{\partial x} + \beta v' = -U\frac{\partial \widehat{h}}{\partial x}, \tag{14}$$

where $\widehat{h} = h_b f_0/H = h_b g/(L_d^2 f_0)$.

Solutions of the form:

$$(\zeta', \psi', \widehat{h}) = \mathrm{Re}\,(\widetilde{\zeta}, \widetilde{\psi}, \widetilde{h}_b)\sin ly\,\mathrm{e}^{\mathrm{i}kx - \omega t} \qquad \text{with} \qquad \omega = \frac{k(UK^2 - \beta)}{K^2 + k_d^2}, \tag{15}$$

Stationary Waves

Stationary solutions when $K = \sqrt{\beta/U}$ and

$$U\frac{\partial \zeta'}{\partial x} + \beta v' = -U\frac{\partial \widehat{h}}{\partial x}, \tag{16}$$

giving

$$\widetilde{\psi} = \mathrm{Re}\,\frac{\widetilde{h}_b}{(K^2 - K_s^2)}. \tag{17}$$

or, with friction

$$\widetilde{\psi} = \mathrm{Re}\,\frac{\widetilde{h}_b}{(K^2 - K_s^2 - \mathrm{i}R)}, \tag{18}$$

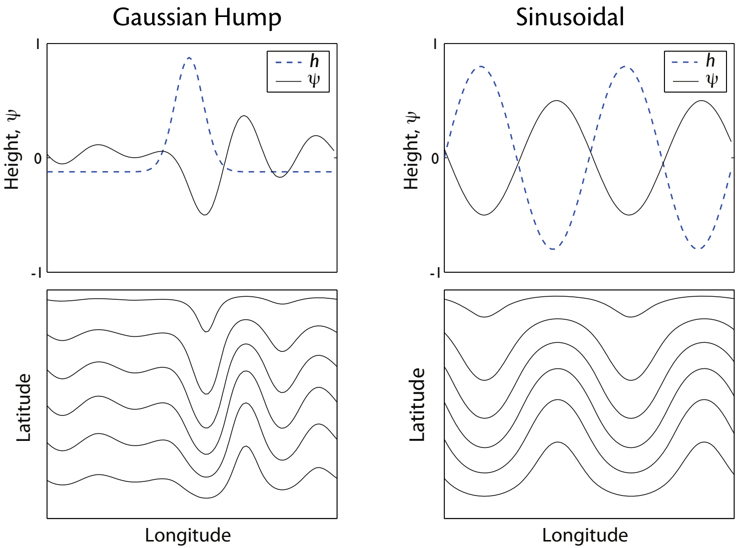
Important Points:

1. There is a resonant response, made finite by friction.
2. Solution can be in various phases with respect to the topography depending on whether $K > K_s$, $K < K_s$ or $K = K_s$ (resonant).
3. Typical for a mountain range to find a minimum in the streamfunction downstream of the ridge.

Some Solutions

Topography and perturbation streamfunction

Streamfunction



Resonant response, a 'trough', just downstream of the ridge.

No resonance. Response out of phase with the topography.

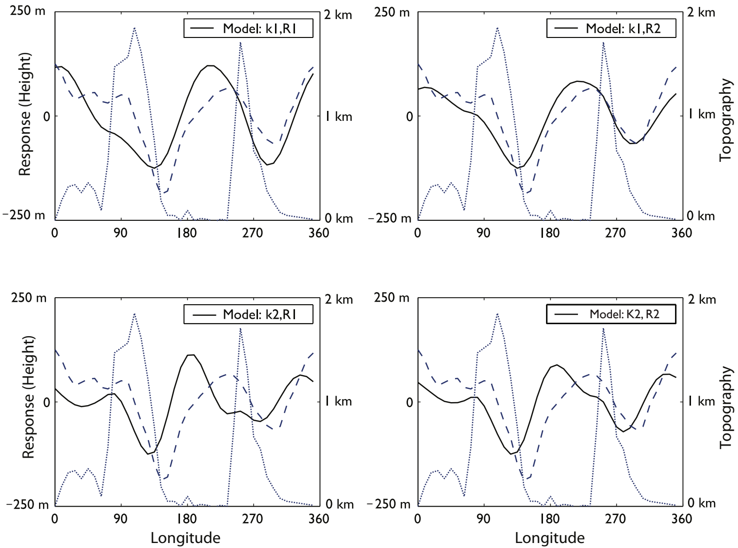
Solutions with Earth Topography

Different Frictional Coefficients and Zonal Winds

Dotted line: Topography.

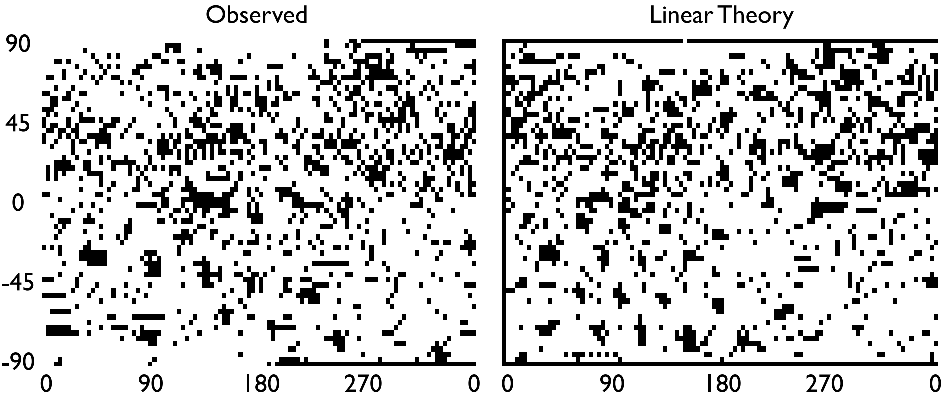
Dashed line: solution.

Solid line: observations



SOLUTIONS USING A GCM

Time averaged streamfunction at 300 hPa.



Comprehensive linear GCM with the observed height-varying, zonally and time-averaged zonal winds.

Note much weaker zonal asymmetries in the southern hemisphere.

Held et al (2002)

Ingredients for Teleconnections

The Story so Far

1. Stationary Rossby waves: Determine background state.
2. Ray theory: tells us where waves from a disturbance go to.

Complications:

- Vertical propagation.
- Nonlinearity.
- Determining the wave source. Thermal effects.

Next Up

(i) Numerical experiments on teleconnections

(ii) The NAO.

Teleconnections and Oscillations

Experiments with GCMs

Geoffrey K. Vallis

Ed Gerber   Stephen Thomson

Beijing, September 2019

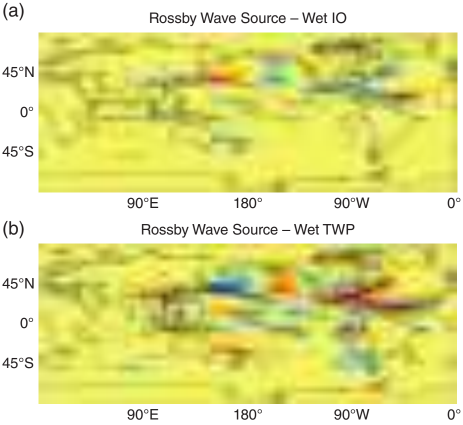
http://tiny.cc/Vallis

Two Topics

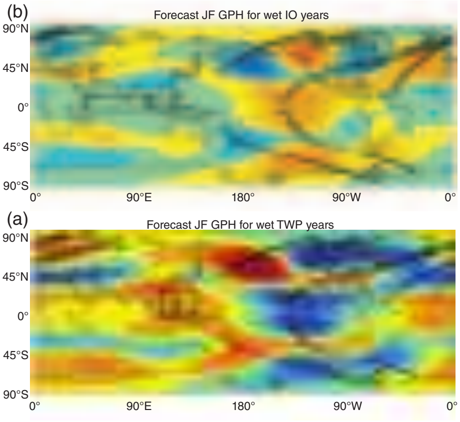
1. Teleconnections from Tropics to Mid-Latitudes
2. The North Atlantic Oscillation and Northern Annular Mode (NAO and NAM).

Rossby Wave Source and Wave Trains

Source: Vortex Stretching & Advection



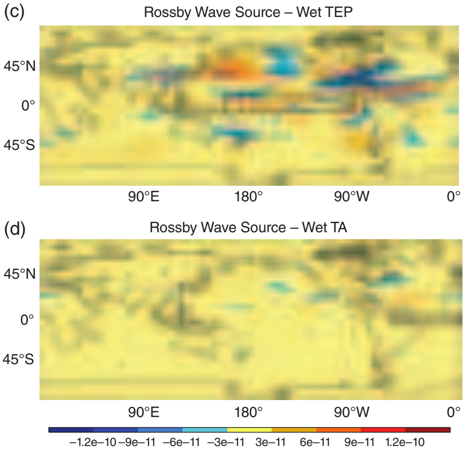
Wave Train



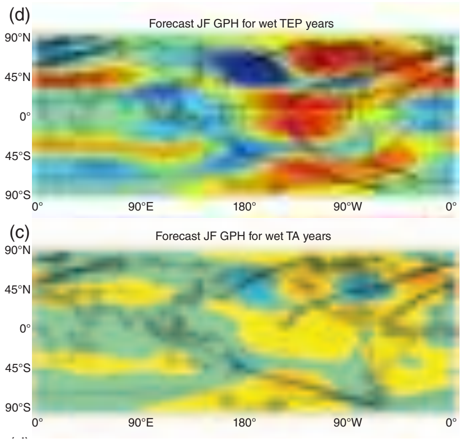
Scaife et al, 2019

Rossby Wave Source and Wave Trains

Source: Vortex Stretching & Advection



Wave Train



Scaife et al, 2019

Goals of the Study

How and why do SST anomalies affect Europe? In particular:

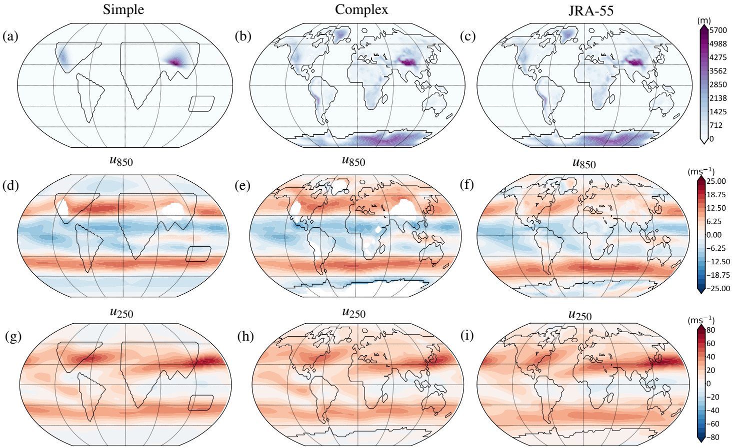
1. Tropical SST anomalies vs mid-latitude SST anomalies.
2. Winter vs summer.

It is generally thought that winter anomalies are more effective than summer ones.
And mid-latitude anomalies less effective than tropical ones. Why?

Experimental Configuration

(i) Impose very localized SST anomalies all over the global (mainly Pacific), in various seasons.

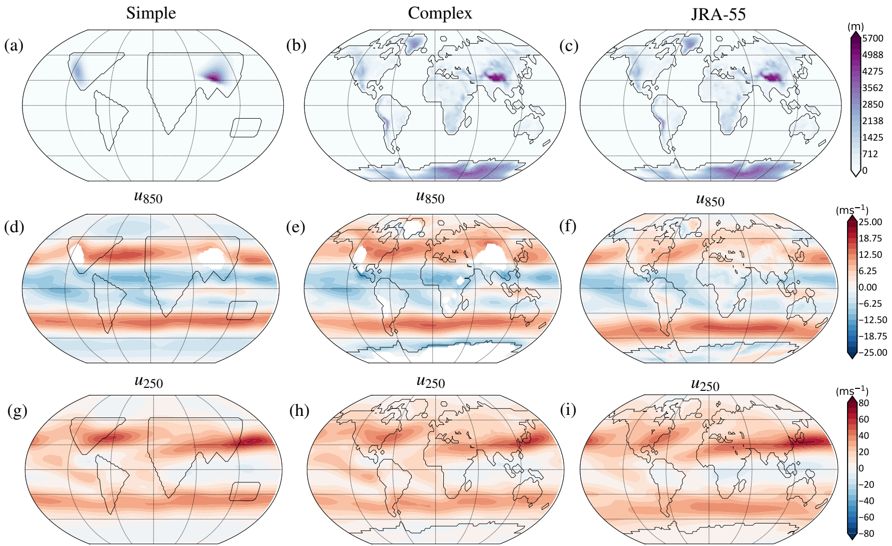
(ii) Vary experimental configuration. (iii) Look for robust responses.



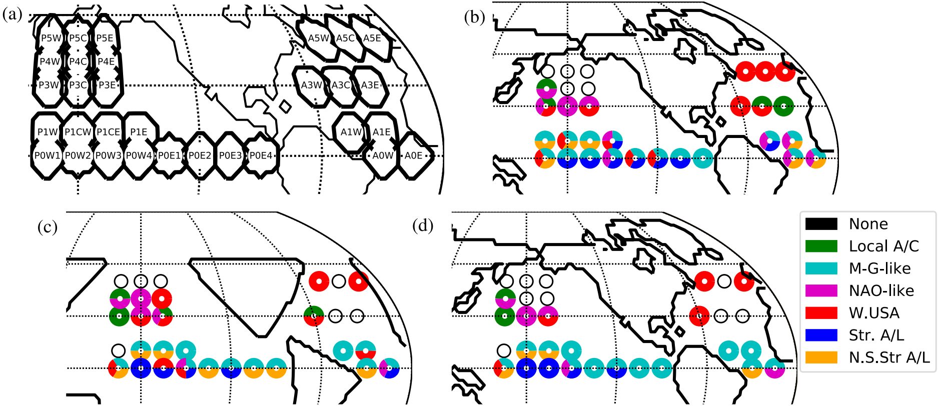
Winter

Note idealized and realistic continental outlines.

Experimental Configuration



Location of SST Anomalies



Each experiment is a single, persistent SST anomaly.

Look at summer and winter responses.

PROCEDURE

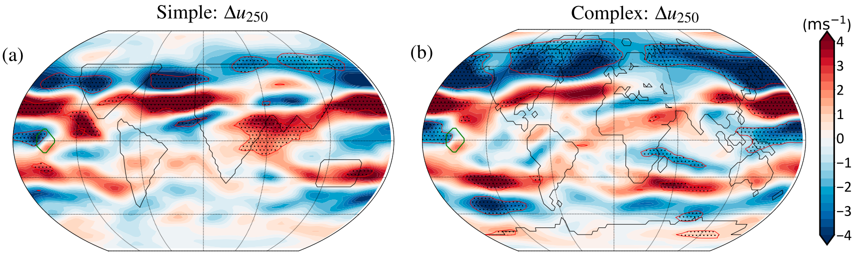
- Construct localized SST (7° wide) anomaly ($\sim 4°$ C) using Q fluxes (so that the SST anomaly arises from anomalous ocean heat transport). 31 locations, in two different configurations.
- Examine response and check for robustness with two conditions
  1. The response in a particular quantity (e.g. 250 hPa zonal wind) must be similar across the two configurations.
  2. The responses within each configuration must be statistically significant, as judged by the t-test with a 95% confidence limit.
  3. Look also at response to neighboring SST anomalies.

Analysis

- **None**
- **Local anticyclonic circulation**: A statistically-significant anticyclone at 250 hPa, indicative of a local linear-like response as in Hoskins (1981).
- **Matsuno-Gill-like**: A statistically-significant response displaying the broad characteristics of the simple linear responses to tropical heating, specifically low-level eastward winds and upper-level westward winds.
- **NAO-like**: A statistically-significant response over the North Atlantic sector that looks like either a positive or negative NAO-like state. For some cases this will constitute a local response, and others (e.g. NAO-like responses to tropical-Pacific SST anomalies) it will constitute a remote teleconnection.
- **Anomaly off the Western US**: A statistically-significant small wind anomaly off the coast of Alaska.
- **Strengthened Aleutian-Low**: A statistically-significant strengthening of the Aleutian-Low in the central North-Pacific.

Example Response

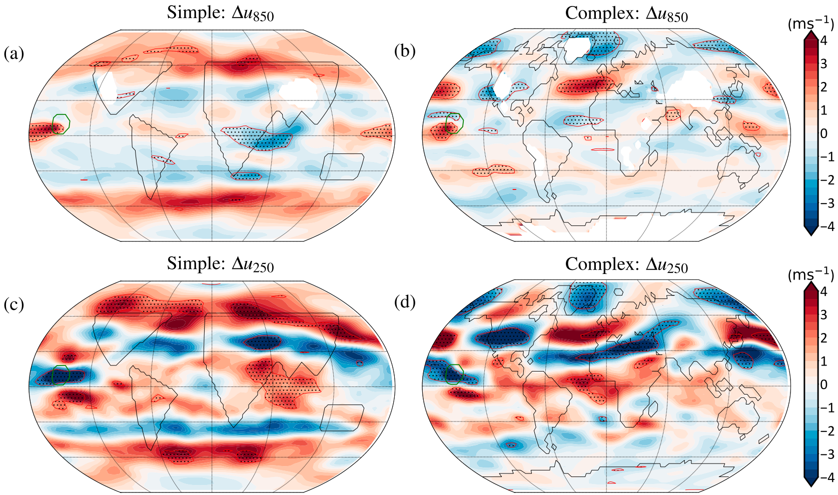
Western equatorial



Response of zonal wind in DJF to SST anomaly in western equatorial Pacific

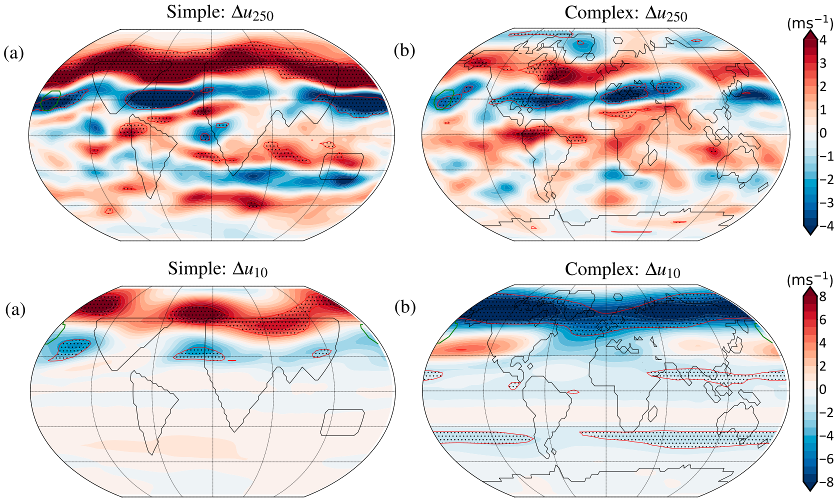
Example Response

Central equatorial



Response of zonal wind in DJF to SST anomaly in central equatorial Pacific

Example Response
Mid and high latitudes



SST anomaly in mid-latitude Pacific

SST anomaly in high latitude Pacific

Results of Analysis

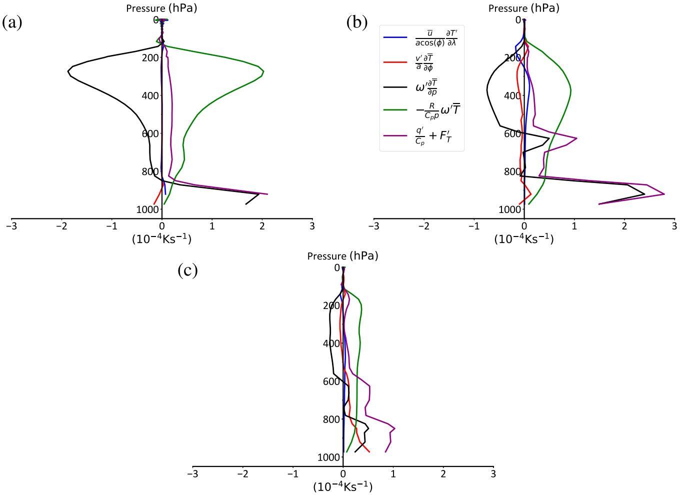
1. Tropical response is ‘robust’, especially SST anomalies in western and central Pacific. Mid-latitude response is weaker and less robust.
2. Winter Response is stronger then summer response.

Why?

Why the Tropics?

SST anomalies reach into the free atmosphere

Decomposition of temperature equation at three locations:



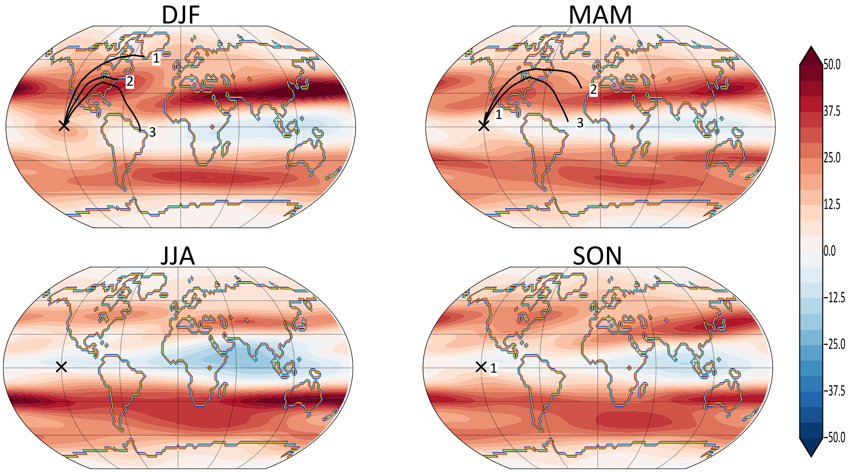
(a) = Central Equatorial Pacific, complex.

(b) = Central Pacific, 25° N, complex

(b) = Central Pacific, 25° N, simple

Only (a) has a local response in the upper atmosphere.

Why Winter not Summer?



Color shading is zonal wind

Rays are trapped in the tropics in summer, but can escape in winter.

Wave propagation requires:

$$0 < U - c < \frac{\beta^*}{k^2}.$$

Need $U > 0$ for stationary waves.

Summary

Mid-latitudes vs Tropics

(i) A tropical SST anomaly can reach the free atmosphere more easily than a mid-latitude anomaly.

(ii) The response to a tropical anomaly is less sensitive to changes in the background climatology than a midlatitude anomaly.

(iii) Mid-latitude SST anomalies exist in a highly variable environment with a low signal-to-noise ratio.

Summer vs Winter

(i) A Rossby ray in winter can potentially propagate from the tropics to the mid-latitudes.

(ii) A Rossby ray in summer is trapped in the tropics by westward winds:

$$l^2(y) = \frac{\beta^*}{U - c} - k^2.$$

So condition for wave propagation is that:

$$0 < U - c < \frac{\beta^*}{k^2}.$$

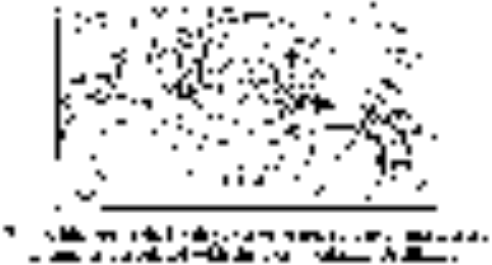
Need $U > 0$.

The Other Kind of Teleconnection

Examples: The NAO/NAM and ENSO

The NAO

> *It has been many times remarked, that the weather in Greenland is just the reverse to that in Europe; so that when the temperate climates are incommoded with a very hard winter, it is here uncommonly mild, and vice versa.*
>
> David Cranzt, The History of Greenland



Questions

1. Why does it exist?
   - Variations of the storm track
2. Why does it change and on what timescales?

The NAO
As an EOF

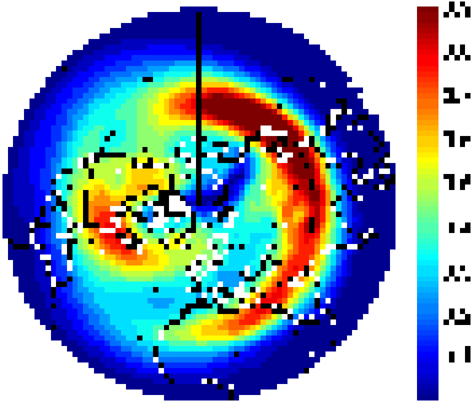
First EOF of hemispheric winter sea level pressure (25% of the variance).

Right: first EOF of winter sea level pressure in the Euro-Atlantic sector only.
(Ambaum et al. 2001.)

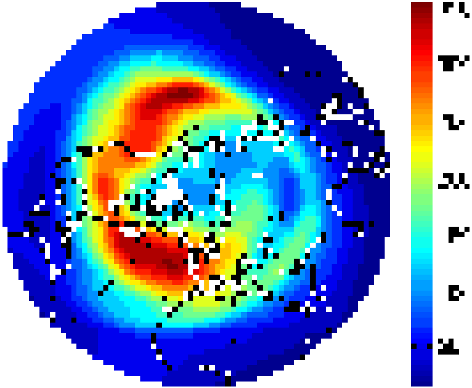
NAO: Eady Growth Rate and Eddy KE

From Observations (re-analysis)

Eady Growth Rate



Eddy Kinetic Energy

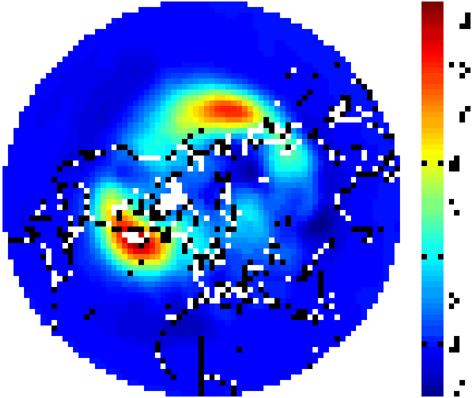


Eddy Kinetic Energy ≈ Storm Track Intensity ≈ co-incident with NAO

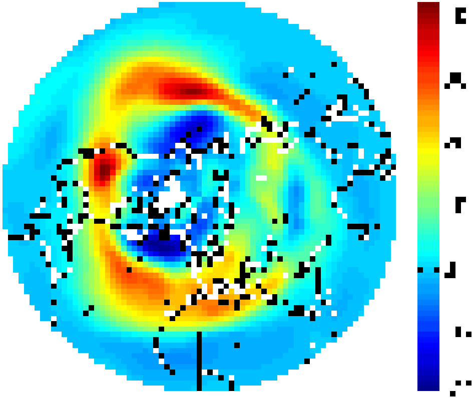
NAO: Eddy Momentum and Heat Fluxes

From Observations (re-analysis)

Eady Heat Flux

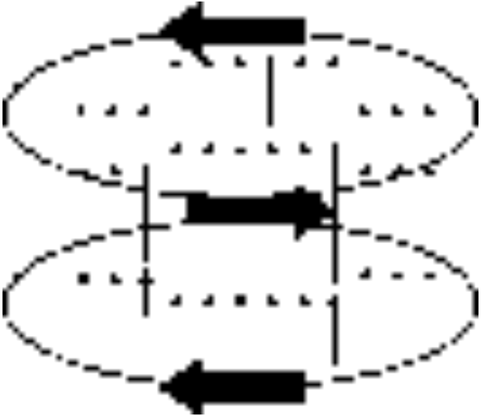


Eddy Momentum Flux (2–10 days)



Eddy momentum fluxes downstream of eddy heat fluxes.

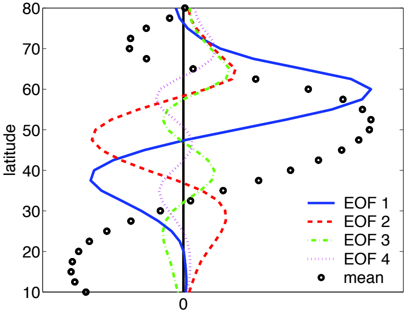
Momentum Fluxes and EOFs

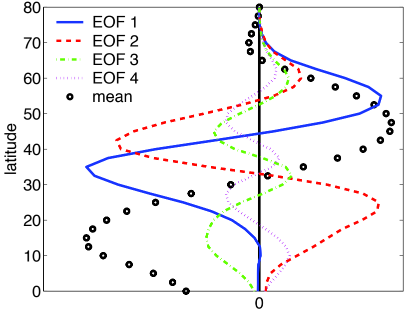


Anomolous momentum fluxes (arrows) give circulation pattern (contours) and hence EOFs.

NAO: Observed EOFs

Observed EOFS

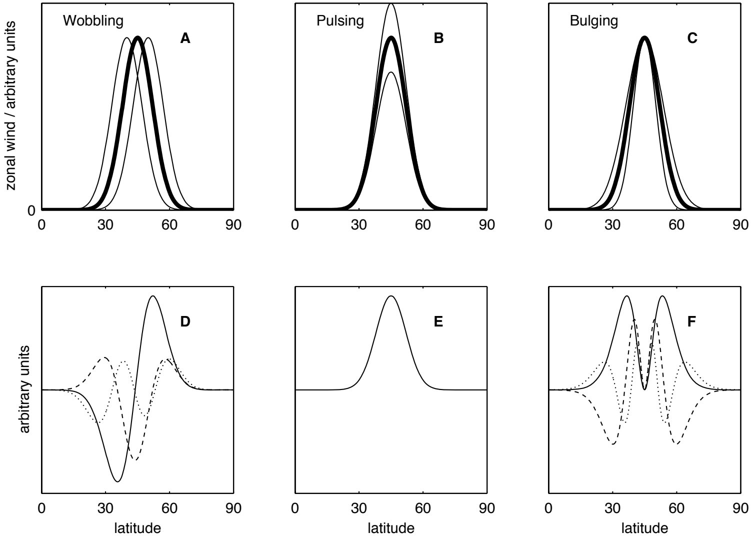




Zonall-averaged zonal wind EOF

Angular momentum EOF.

NAO - Jets and EOFs



Jet variability

Associated EOFs

Wittman et al 2005

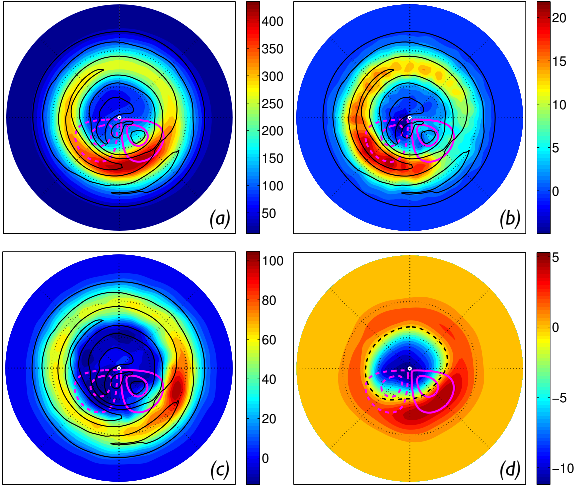
Localized Eddy Fluxes

The NAO and NAM

Add a mountain or a thermal anomaly and create a storm track.

EKE

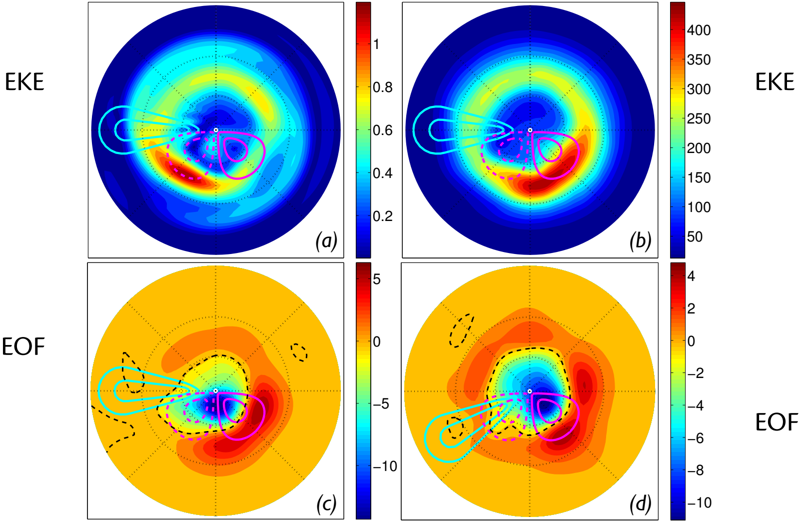
Eddy Heat Fluxes



Eddy Momentum Fluxes

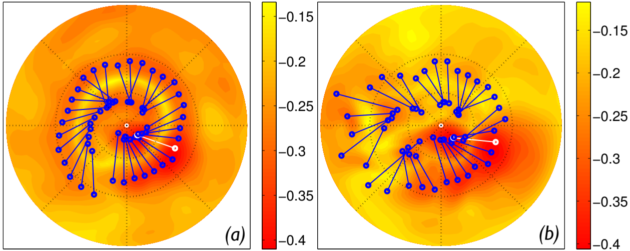
EOF

NAO - Eddy Fluxes and EOFs

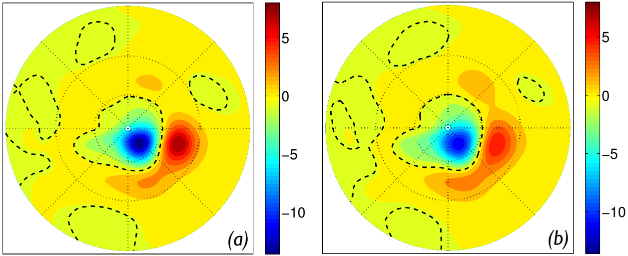


NAO - Teleconnections

Teleconnection Lines



Teleconnection Patterns



Regress the pressure field onto the timeseries of pressure difference between the two most anticorrelated pairs.

$$T_c(t) = A\left[\frac{p'(\boldsymbol{x}_1, t)}{\sigma_{\boldsymbol{x}_1}} - \frac{p'(\boldsymbol{x}_2, t)}{\sigma_{\boldsymbol{x}_2}}\right]$$

$$P_c(\boldsymbol{x}) = \frac{1}{n}\sum_{t=0}^{t=n} p'(\boldsymbol{x}, t)T_c(t).$$

Some Conclusions

- The NAO is a see-saw type teleconnetction
- Dipolar pattern – consequence of one baroclinic zone.
- Two phases: (i) Eddy momentum fluxes strong, mid-latitude jet and sub-tropical jet become distinct. (ii) Eddy momentum fluxes are weak subtropics, the mid-latitude jet is weak and the sub-tropical jet dominates.
- The NAO is the local manifestation of hemispheric dynamics (still a topic of debate).

Overall Conclusions

Teleconnections: Arise from a variety of sources and mechanisms.

1. Organized variability gives anti-correlations ('see-saws').
    - The NAO, NAM and SAM.
        - Storm track variability on synoptic timescales. SST variability can give year-to-year variability.
        - Stationary waves organize patterns into NAO.
        - Zonally-symmetric hemispheric-wide version is the SAM (and NAM).
    - ENSO (did not discuss that).
2. Long distance teleconnections, for example from tropics to mid-latitudes.
    - Rossby wave packets (ray tracing).
    - Stratospheric bridge.
3. Other mechanisms you will hear about this week!